

# Differentiation in Trigonometry — Topics and Subtopics

Derivatives of the trigonometric functions · Grade 11 · Pure Mathematics 1, 2 and 3

206 subtopics — FOUNDATIONS 31 · THE DERIVATIVES 37 · THE RULES APPLIED 62 · APPLICATIONS 64 · ERRORS AND CHECKS 12

## PART A — FOUNDATIONS

### A.1 Angle measure

- A.1.1 Degree measure
- A.1.2 Radian measure
- A.1.3 Conversion between degrees and radians
- A.1.4 Why the derivatives require radians
- A.1.5 Derivative of  $\sin x^\circ$  as  $(\pi/180) \cdot \cos x^\circ$
- A.1.6 Exact values of the ratios in radians
- A.1.7 The unit circle definition of the ratios
- A.1.8 Periodicity of each ratio
- A.1.9 Domain of each ratio

### A.2 Identities required

- A.2.1  $\sin^2 x + \cos^2 x = 1$
- A.2.2  $\tan x = \sin x / \cos x$
- A.2.3  $\cot x = \cos x / \sin x$
- A.2.4  $\sec x = 1 / \cos x$
- A.2.5  $\operatorname{cosec} x = 1 / \sin x$
- A.2.6  $1 + \tan^2 x = \sec^2 x$

- A.2.7  $1 + \cot^2 x = \operatorname{cosec}^2 x$
- A.2.8 Addition formulae for  $\sin$ ,  $\cos$  and  $\tan$
- A.2.9 Double angle formulae
- A.2.10 Half angle formulae
- A.2.11 Power reduction formulae
- A.2.12 Product to sum formulae
- A.2.13 Sum to product formulae
- A.2.14 The  $R \sin(x \pm \alpha)$  and  $R \cos(x \pm \alpha)$  forms

### A.3 Limits required

- A.3.1  $\lim_{h \rightarrow 0} \sin h / h = 1$
- A.3.2 Geometric proof of  $\lim \sin h / h$
- A.3.3 The squeeze theorem applied to  $\sin h / h$
- A.3.4  $\lim_{h \rightarrow 0} (1 - \cos h)/h = 0$
- A.3.5  $\lim_{h \rightarrow 0} (1 - \cos h)/h^2 = 1/2$
- A.3.6  $\lim_{h \rightarrow 0} \tan h / h = 1$
- A.3.7 Continuity of the trigonometric functions
- A.3.8 Points of discontinuity of  $\tan$ ,  $\cot$ ,  $\sec$  and  $\operatorname{cosec}$

## PART B — THE DERIVATIVES

### B.1 From first principles

- B.1.1 The difference quotient for  $\sin x$
- B.1.2 Expanding  $\sin(x + h)$  by the addition formula
- B.1.3 Applying the two standard limits
- B.1.4 Derivative of  $\sin x$  equals  $\cos x$
- B.1.5 The difference quotient for  $\cos x$
- B.1.6 Derivative of  $\cos x$  equals  $-\sin x$
- B.1.7 Derivative of  $\tan x$  from first principles
- B.1.8 Derivative of  $\tan x$  by the quotient rule
- B.1.9 Geometric interpretation on the unit circle
- B.1.10 The cycle  $\sin \rightarrow \cos \rightarrow -\sin \rightarrow -\cos$

### B.2 The six circular functions

- B.2.1  $d/dx (\sin x) = \cos x$
- B.2.2  $d/dx (\cos x) = -\sin x$
- B.2.3  $d/dx (\tan x) = \sec^2 x$
- B.2.4  $d/dx (\cot x) = -\operatorname{cosec}^2 x$
- B.2.5  $d/dx (\sec x) = \sec x \cdot \tan x$
- B.2.6  $d/dx (\operatorname{cosec} x) = -\operatorname{cosec} x \cdot \cot x$
- B.2.7 Domain on which each derivative is valid
- B.2.8 Points at which  $\tan$  and  $\sec$  are not differentiable
- B.2.9 The minus sign on every co-function

- B.2.11 Learning the table by the co-function pattern

### B.3 Inverse trigonometric functions

- B.3.1 Restricted domains that make the inverses exist
- B.3.2  $d/dx (\arcsin x) = 1/\sqrt{1 - x^2}$
- B.3.3  $d/dx (\arccos x) = -1/\sqrt{1 - x^2}$
- B.3.4  $d/dx (\arctan x) = 1/(1 + x^2)$
- B.3.5  $d/dx (\operatorname{arccot} x) = -1/(1 + x^2)$
- B.3.6  $d/dx (\operatorname{arcsec} x)$
- B.3.7  $d/dx (\operatorname{arccosec} x)$
- B.3.8 Derivation by implicit differentiation
- B.3.9 Derivation by the inverse function rule
- B.3.10 Derivatives of  $\arcsin(x/a)$  and  $\arctan(x/a)$
- B.3.11 Why the derivatives of  $\arcsin$  and  $\arccos$  sum to zero

### B.4 Hyperbolic functions

- B.4.1  $d/dx (\sinh x) = \cosh x$
- B.4.2  $d/dx (\cosh x) = \sinh x$
- B.4.3  $d/dx (\tanh x) = \operatorname{sech}^2 x$
- B.4.4 Sign patterns compared with the circular functions
- B.4.5 Derivatives of the inverse hyperbolic functions

## PART C — THE RULES APPLIED

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### C.1 The chain rule

- C.1.1  $d/dx \sin(ax + b)$
- C.1.2  $d/dx \cos(ax + b)$
- C.1.3  $d/dx \tan(ax + b)$
- C.1.4  $d/dx \sin(f(x))$
- C.1.5  $d/dx \sin^n x$
- C.1.6  $d/dx \cos^n x$
- C.1.7  $d/dx \tan^n x$
- C.1.8 Distinguishing  $\sin^2 x$  from  $\sin(x^2)$
- C.1.9  $d/dx \sin(x^2)$
- C.1.10 Chains of three links such as  $\sin^3(2x + 1)$
- C.1.11  $d/dx \sqrt{\sin x}$
- C.1.12  $d/dx e^{\sin x}$
- C.1.13  $d/dx \ln(\sin x)$
- C.1.14  $d/dx \ln(\sec x + \tan x)$
- C.1.15  $d/dx \sin(\ln x)$
- C.1.16  $d/dx \arctan(f(x))$

### C.2 The product rule

- C.2.1  $d/dx (x \cdot \sin x)$
- C.2.2  $d/dx (x^2 \cdot \cos x)$
- C.2.3  $d/dx (e^x \cdot \sin x)$
- C.2.4  $d/dx (\sin x \cdot \cos x)$
- C.2.5 Simplifying by a double angle identity first
- C.2.6 Products of three factors
- C.2.7 Product combined with a chain

### C.3 The quotient rule

- C.3.1  $d/dx (\sin x / x)$
- C.3.2  $d/dx (\sin x / \cos x)$  giving  $\sec^2 x$
- C.3.3  $d/dx (1 / \sin x)$  giving  $-\operatorname{cosec} x \cot x$
- C.3.4  $d/dx ((1 + \sin x)/(1 + \cos x))$
- C.3.5  $d/dx (x / \tan x)$
- C.3.6 Simplifying the result with a Pythagorean identity
- C.3.7 Choosing the product rule in place of the quotient rule

### C.4 Combined rules

- C.4.1 A chain inside a product

- C.4.2 A chain inside a quotient
- C.4.3 A product inside a chain
- C.4.4 Order of application
- C.4.5 Factorising the answer
- C.4.6 Simplifying the answer with identities

### C.5 Implicit differentiation

- C.5.1 Differentiating  $\sin y$  with respect to  $x$
- C.5.2 Differentiating  $\cos(xy)$
- C.5.3 Curves such as  $\sin x + \cos y = 1$
- C.5.4 Curves such as  $x \cdot \sin y = y \cdot \cos x$
- C.5.5 Tangents to an implicit trigonometric curve
- C.5.6 Deriving the inverse trigonometric derivatives

### C.6 Parametric differentiation

- C.6.1 The circle  $x = a \cos t$ ,  $y = a \sin t$
- C.6.2 The ellipse  $x = a \cos t$ ,  $y = b \sin t$
- C.6.3 The astroid  $x = a \cos^3 t$ ,  $y = a \sin^3 t$
- C.6.4 The cycloid  $x = t - \sin t$ ,  $y = 1 - \cos t$
- C.6.5  $dy/dx$  as  $(dy/dt) \div (dx/dt)$
- C.6.6 Second derivative of a parametric curve
- C.6.7 Tangents and normals to a parametric curve
- C.6.8 Stationary points of a parametric curve

### C.7 Logarithmic differentiation

- C.7.1  $y = (\sin x)^x$
- C.7.2  $y = x^{\sin x}$
- C.7.3  $y = (\tan x)^{\cos x}$
- C.7.4 Long products of trigonometric factors

### C.8 Higher derivatives

- C.8.1 Second derivative of  $\sin x$  and  $\cos x$
- C.8.2 The four-step cycle of repeated differentiation
- C.8.3  $n$ th derivative of  $\sin x$  as  $\sin(x + n\pi/2)$
- C.8.4  $n$ th derivative of  $\cos x$  as  $\cos(x + n\pi/2)$
- C.8.5  $n$ th derivative of  $\sin(ax + b)$
- C.8.6 The equation  $y'' + y = 0$
- C.8.7 The equation  $y'' + \omega^2 y = 0$
- C.8.8 Leibniz rule applied to  $x \cdot \sin x$

## PART D — APPLICATIONS

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### D.1 Tangents and normals

- D.1.1 Gradient of a trigonometric curve at a point
- D.1.2 Equation of the tangent
- D.1.3 Equation of the normal
- D.1.4 Horizontal tangents of  $y = \sin x$
- D.1.5 Points with a given gradient
- D.1.6 Tangent at the origin to  $y = \sin x$
- D.1.7 Tangent at the origin to  $y = \tan x$
- D.1.8 Angle of intersection of two trigonometric curves

### D.2 Behaviour of a trigonometric function

- D.2.1 Intervals of increase and decrease

- D.2.2 Stationary points of  $y = \sin x$  and  $y = \cos x$
- D.2.3 Stationary points of  $y = a \sin x + b \cos x$
- D.2.4 The first derivative test
- D.2.5 The second derivative test
- D.2.6 Concavity of a trigonometric curve
- D.2.7 Points of inflection
- D.2.8 Sketching  $y = x + \sin x$
- D.2.9 Sketching  $y = \sin x \cdot \cos x$
- D.2.10 Sketching  $y = e^{(-x)} \cdot \sin x$
- D.2.11 Repeating stationary points and periodicity
- D.2.12 Greatest and least value on a closed interval
- D.2.13 Maximum of  $a \sin x + b \cos x$  by the R form

D.2.14 The R form compared with the calculus method

### D.3 Optimisation

- D.3.1 An objective function written in terms of an angle
- D.3.2 The feasible interval for an angle
- D.3.3 Largest rectangle inscribed in a circle
- D.3.4 Largest rectangle inscribed in a semicircle
- D.3.5 Trough of greatest cross-sectional area
- D.3.6 Ladder carried round a corner
- D.3.7 Greatest area of a triangle with a fixed side
- D.3.8 Shortest crossing path
- D.3.9 Refraction and the law of least time
- D.3.10 Greatest volume of a cone made from a sector
- D.3.11 Greatest range of a projectile
- D.3.12 Justifying the maximum or the minimum

### D.4 Rates of change and motion

- D.4.1 Angular velocity  $d\theta/dt$
- D.4.2 Connected rates with a rotating rod
- D.4.3 Lengthening shadow from a moving light
- D.4.4 Rotating searchlight on a straight shore
- D.4.5 Sliding ladder in terms of the angle
- D.4.6 Simple harmonic motion  $x = a \sin(\omega t + \phi)$
- D.4.7 Velocity and acceleration in simple harmonic motion
- D.4.8 The equation of motion  $\ddot{x} = -\omega^2 x$
- D.4.9 Amplitude, period and phase

D.4.10 Oscillation of a pendulum

D.4.11 Alternating current and voltage

D.4.12 Tidal and seasonal models

### D.5 Approximation and series

- D.5.1 Small angle approximation  $\sin x \approx x$
- D.5.2 Small angle approximation  $\cos x \approx 1 - x^2/2$
- D.5.3 Small angle approximation  $\tan x \approx x$
- D.5.4 Validity of the small angle approximations
- D.5.5 Linear approximation from the derivative
- D.5.6 Estimating  $\sin 31^\circ$  and similar values
- D.5.7 Percentage error in a trigonometric quantity
- D.5.8 Maclaurin series for  $\sin x$
- D.5.9 Maclaurin series for  $\cos x$
- D.5.10 Maclaurin series for  $\tan x$
- D.5.11 Maclaurin series for  $\arctan x$
- D.5.12 Newton-Raphson applied to a trigonometric equation

### D.6 The reverse direction

- D.6.1 Each derivative read backwards as an integral
- D.6.2  $\int \sin x \, dx$  and the change of sign
- D.6.3  $\int \sec^2 x \, dx$
- D.6.4  $\int \sec x \cdot \tan x \, dx$
- D.6.5 Recognising  $f'(x)$  inside a trigonometric integrand
- D.6.6 Checking a trigonometric integral by differentiating

## PART E — ERRORS AND CHECKS

### E.1 Common errors and checks

- E.1.1 Working in degrees instead of radians
- E.1.2 Losing the minus sign on the derivative of  $\cos x$
- E.1.3 Confusing  $\sin^2 x$  with  $\sin(x^2)$
- E.1.4 Omitting the inner derivative in the chain rule
- E.1.5 Omitting the factor  $a$  in  $\sin(ax + b)$
- E.1.6 Reading  $\sin^{-1} x$  as  $1/\sin x$

- E.1.7 Ignoring the domain restriction of an inverse function
- E.1.8 Differentiating at a point where the function is undefined
- E.1.9 Leaving an answer unsimplified by an identity
- E.1.10 Checking a derivative numerically
- E.1.11 Checking a derivative at a known point
- E.1.12 Checking by integrating back